Pertemuan 9

Date Selasa / 24 April 2024

Tentukan $q(t)$ steady state (ss) bila pada $t=0$, $q=10$, $q'=0$ dan sketsa grafik dengan geogebra

1 $q'' + 4q' + 3q = 15 \sin 2t$

Pers. Karakteristik:

$m^2 + 4m + 3 = 0$

$(m+3)(m+1) = 0$

$m = -3 \quad \vee \quad m = -1$

$q_h(t) = C_1 e^{-3t} + C_2 e^{-t}$

$f(t) = 15 \sin 2t$

$v(t) = a \sin 2t + b \cos 2t$

$v'(t) = 2a \cos 2t - 2b \sin 2t$

$v''(t) = -4a \sin 2t - 4b \cos 2t$

$-4a \sin 2t - 4b \cos 2t + 8a \cos 2t - 8b \sin 2t + 3a \sin 2t + 3b \cos 2t = 15 \sin 2t$

$(-4a - 8b + 3a) \sin 2t + (-4b + 8a + 3b) \cos 2t = 15 \sin 2t$

$-4a - 8b + 3a = 15$

$-a - 8b = 15 \ldots (1)$

$-4b + 8a + 3b = 0$

$8a - b = 0$

(1) dan (2)

$-a - 8b = 15 \quad | \times 1 \longrightarrow -a - 8b = 15$

$8a - b = 0 \quad | \times 8 \quad 64a - 8b = 0 \; -$

$-65a = 15$

$a = \frac{-15}{65} = \frac{-3}{13} \qquad b = -\frac{24}{13}$

Solusi PDNH

$q(t) = C_1 e^{-3t} + C_2 e^{-t} + \frac{3}{13} \sin 2t - \frac{24}{13} \cos 2t$

$q'(t) = -3C_1 e^{-3t} - C_2 e^{-3t} - \frac{6}{13} \cos 2t + \frac{48}{13} \sin 2t$

$q(0) = 10$

$C_1 + C_2 - \frac{24}{13} = 10$

$C_1 + C_2 = \frac{154}{13} \quad \dots (1)$

$q'(0) = 0$

$-3C_1 - C_2 - \frac{K}{13} = 0$

$-3C_1 - C_2 = \frac{16}{13}$

(1) dan (2)

$C_1 + C_2 = {}^{154}/_{13}$

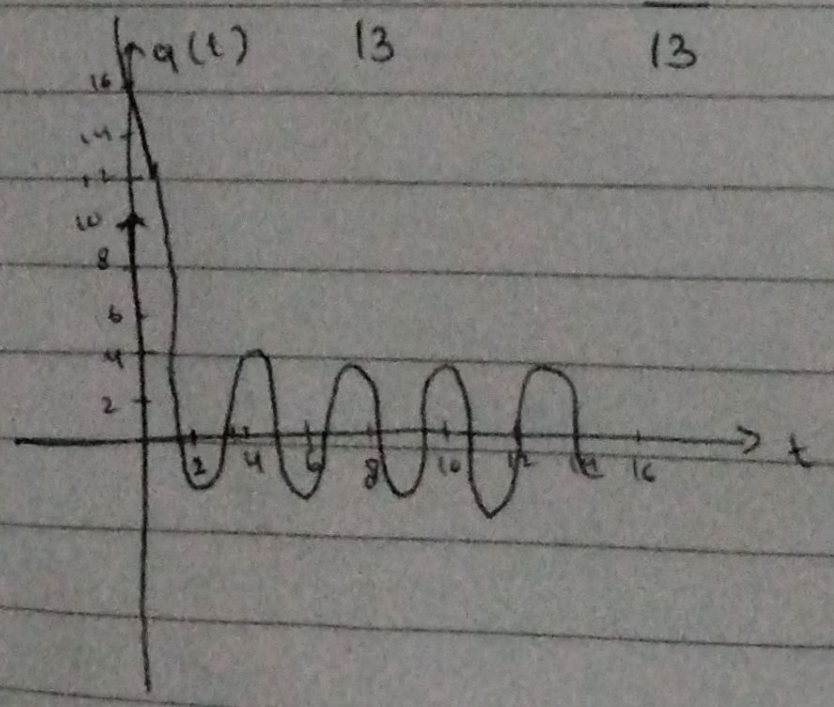
$-3C_1 - C_2 = {}^{6}/_{13} \; +$

$-2C_1 = \frac{160}{13}$

$C_1 = -\frac{160}{26} = -\frac{80}{13}$ $\qquad C_2 = 18$

$$q(t) = -\frac{80}{13}e^{-3t} + 18e^{-t} - \frac{3}{13}\sin 2t - \frac{24}{13}\cos 2t \; //$$

$q(t)$ steady state $= -\frac{3}{13}\sin 2t - \frac{24}{13}\cos 2t$ //



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2. $q'' + 4q' + 4q = 15 \sin 2t$

Pers. Karakteristik:

$m^2 + 4m + 4 = 0$ $\qquad q_{(h)}(t) = e^{-2t}(C_1 + C_2 t)$

$(m+2)(m+2) = 0$

$m = -2 \vee m = -2$

$f(t) = 15 \sin 2t$

$v(t) = a \sin 2t + b \cos 2t$

$v'(t) = 2a \cos 2t - 2b \sin 2t$

$v''(t) = -4a \sin 2t - 4b \cos 2t$

$(-4a - 8b + 4a) \sin 2t + (-4b + 8a + 4b) \cos 2t = 15 \sin 2t$

$-4a - 8b + 4a = 15$ $\qquad -4b + 8a + 4b = 0$

$-8b = 15$ $\qquad 8a = 0$

$b = -\frac{15}{8}$ ... (1) $\qquad a = 0$ ... (2)

$q(t) = q_h(t) + v(t)$

$= e^{-2t}(C_1 + C_2 t) - \frac{15}{8} \cos 2t$

$q(t)$ steady state $= -\frac{15}{8} \cos 2t$ //

$q'(t) = -2e^{-2t}(C_1 + C_2 t) + e^{-2t} C_2 + \frac{30}{8} \sin 2t$

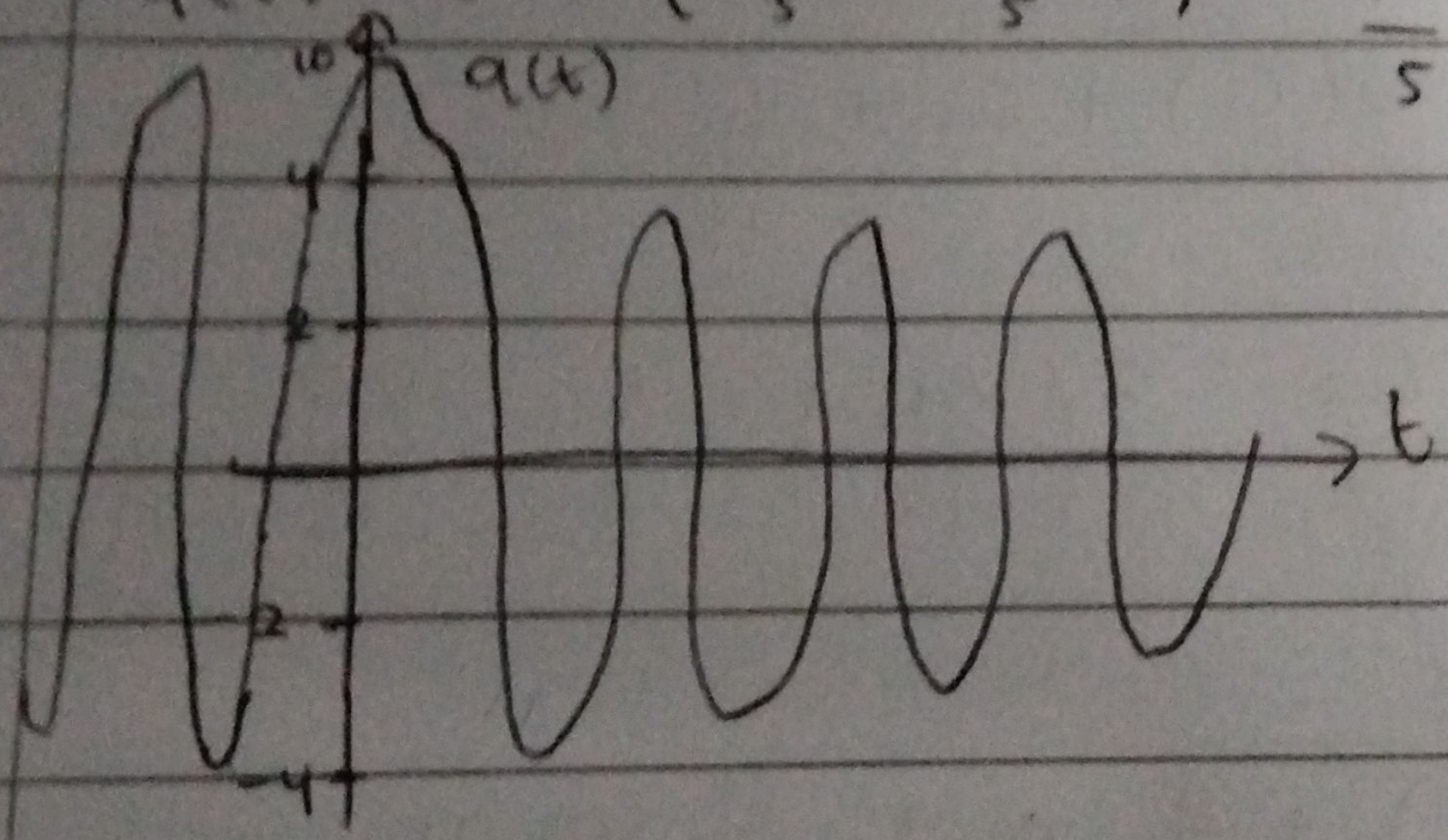
$= -2C_1 e^{-2t} - 2C_2 e^{-2t} t + C_2 e^{-2t} + \frac{30}{8} \sin 2t$

$q(0) = 10$ $\qquad q'(0) = 0$

$C_1 - \frac{15}{8} = 10 \rightarrow C_1 = \frac{15}{8} + 10$ $\qquad -2C_1 + C_2 = 0$

$\boxed{C_1 = \frac{95}{8}}$ $\qquad \boxed{C_2 = 2C_1 = 2\left(\frac{95}{8}\right) = \frac{190}{8}}$

$$q(t) = e^{-2t}\left(\frac{60}{5} + \frac{136}{5}t\right) - \frac{18}{5}\cos 2t$$



3 $q'' + 4q' + 5q = 15 \sin 2t$

Pers Karakteristik:

$m^2 + 4m + 5 = 0$

$m_{(1,2)} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-4 \pm \sqrt{16 - 20}}{2} = \frac{-2 \pm \sqrt{-4}}{2} = -2 \pm j$

$q_h(t) = e^{-2t}(C_1 \cos t + C_2 \sin t)$

$f(t) = 15 \sin 2t$

$v(t) = a \sin 2t + b \cos 2t$

$v'(t) = 2a \cos 2t - 2b \sin 2t$

$v''(t) = -4a \sin 2t - 4b \cos 2t$

$(-4a - 8b + 5a) \sin 2t + (-4b + 8a + 5b) \cos 2t = 15 \sin 2t$

$-4a - 8b + 5a = 15$ $\qquad -4b + 8a + 5b = 0$

$a - 8b = 15 \ldots (1)$ $\qquad 8a + b = 0 \ldots (2)$

(1) dan (2)

$a - 8b = 15$	$\times 1$	$a - 8b = 15$
$8a + b = 0$	$\times 8$	$64a + 8b = 0$ +
		$65a = 15$

$a = \frac{15}{65} = \frac{3}{13}$ $\qquad b = -\frac{24}{13}$

$q(t) = q_h(t) + v(t)$

$= e^{-2t}(C_1 \cos t + C_2 \sin t) + \frac{3}{13} \sin 2t - \frac{24}{13} \cos 2t$

$q(t)$ steady state $= \frac{3}{13} \sin 2t - \frac{24}{13} \cos 2t$ //

$q'(t) = -2e^{-2t}(C_1 \cos t + C_2 \sin t) + e^{-2t}(-C_1 \sin t + C_2 \cos t) + \frac{6}{13} \cos 2t + \frac{48}{13} \sin 2t$

$q(0) = 10$

$C_1 - \frac{24}{13} = 10 \rightarrow C_1 = \frac{154}{13}$

$q'(0) = 0$

$-2(C_1) + C_2 + \frac{6}{13} = 0$

$C_2 = 2\left(\frac{154}{13}\right) - \frac{6}{13} = \frac{308}{13} - \frac{6}{13} = \frac{302}{13}$

$q(t) = e^{-2t}\left(\frac{154}{13}\cos t + \frac{302}{13}\sin t\right) + \frac{3}{13}\sin 2t - \frac{24}{13}\cos 2t$

